

Masked Multi-Head Self-Attention in the Transformer Decoder

Convention: 1 = ON / allowed, 0 = OFF / blocked. Combine the two binary masks element-wise, then convert 1 to 0 and 0 to negative infinity before Softmax.

Core idea: causal masking controls future-token visibility; padding masking controls meaningless PAD positions.

1. Position in the Decoder

```
Target tokens
|
Embedding + Positional Encoding
|
Masked Multi-Head Self-Attention
|
Add + LayerNorm
|
Cross Attention
|
Add + LayerNorm
|
Feed Forward Network
|
Add + LayerNorm
```

2. Why Masking Is Needed

For **I am a student**, the decoder training input is shifted right:

<BOS> I am a student

While processing a position, the decoder must not use future target tokens. For example, the position **am** can use BOS, I and am, but not a or student. This prevents future-information leakage.

3. Create Q, K and V

Because this is **self-attention**, Q, K and V all come from the decoder input X:

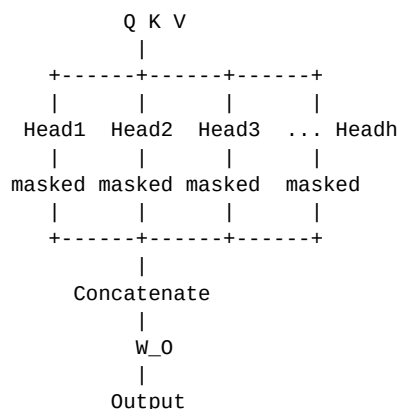
$$\begin{aligned}Q &= XW_Q \\K &= XW_K \\V &= XW_V\end{aligned}$$

Scaled attention scores:

$$\text{Scores} = QK^T / \text{sqrt}(d_k)$$

4. Multi-Head Attention

The representation is divided across multiple heads. Example: $d_{\text{model}} = 512$ with 8 heads gives $d_k = d_v = 64$.



5. Mask 1: Causal / Look-Ahead Mask

For **BOS, I, am, a, student**, use 1 = ON and 0 = OFF:

Query / Key	BOS	I	am	a	student
BOS	1	0	0	0	0
I	1	1	0	0	0
am	1	1	1	0	0
a	1	1	1	1	0
student	1	1	1	1	1

A row is one Query and a column is one Key. The lower-triangular pattern means each token can attend to itself and earlier tokens, but not future tokens.

6. Mask 2: Padding Mask

Suppose the sequence contains padding:

<BOS> I am a student <PAD> <PAD>

The padding mask marks real key positions as 1 and PAD key positions as 0:

Key	BOS	I	am	a	student	PAD	PAD
Padding mask	1	1	1	1	1	0	0

7. Combine Both Masks by Element-wise Multiplication

FinalMask = CausalMask x PaddingMask

Causal	Padding	Final	Meaning
1	1	1	Allowed by both
1	0	0	Padding blocks it
0	1	0	Causal rule blocks it
0	0	0	Both block it

Thus, a position remains ON only when **both** masks are ON. For binary masks this is equivalent to a logical AND.

8. Mask Multiplication Example

Causal mask:

```
1 0 0 0 0
1 1 0 0 0
1 1 1 0 0
1 1 1 1 0
1 1 1 1 1
```

Padding mask, where the last position is PAD:

```
1 1 1 1 0
1 1 1 1 0
1 1 1 1 0
1 1 1 1 0
1 1 1 1 0
```

Element-wise product:

```
1 0 0 0 0
1 1 0 0 0
1 1 1 0 0
1 1 1 1 0
1 1 1 1 0
```

9. Convert the Binary Mask

After obtaining the final binary mask:

1 -> 0

0 -> -infinity

Allowed positions get 0 so their scores remain unchanged. Blocked positions get negative infinity so their Softmax probability becomes zero.

10. Exact Location of the Mask

```
Q
|
K
|
QK^T
|
Divide by sqrt(d_k)
|
Scaled Attention Scores
|
+ Mask
|
Softmax
|
Attention Weights
|
multiply by V
```

$$\text{Attention}(Q,K,V) = \text{Softmax}(QK^T / \sqrt{d_k} + \text{Mask}) V$$

The mask changes the **attention scores**; it does not directly delete Q, K or V.

11. Numerical Example

Suppose one row of scaled scores is:

2.1 1.4 3.2 0.8 2.0

Mask for that row:

0 0 -infinity -infinity -infinity

After adding the mask:

2.1 1.4 -infinity -infinity -infinity

Since $\exp(-\text{infinity}) = 0$, the blocked positions receive zero probability after Softmax. The remaining probabilities depend on the original scores.

12. Why Negative Infinity?

Simply adding 0 to a blocked score would not block it because Softmax would still assign it some probability. Using negative infinity makes its exponential contribution zero.

Blocked score:
 $\text{score} + (-\text{infinity}) = -\text{infinity}$

Softmax:
 $\exp(-\text{infinity}) = 0$

13. Multiply the Attention Weights by V

After Softmax:

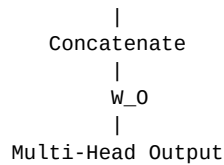
$$\text{Output} = \text{AttentionWeights} \times V$$

Example:
 $0.10 * V(\text{BOS})$
 $+ 0.25 * V(\text{I})$
 $+ 0.65 * V(\text{am})$
 $+ 0 * V(\text{a})$
 $+ 0 * V(\text{student})$

Future or padded positions have zero weight, so their V vectors contribute nothing.

14. What Happens Across Heads?

Head 1 -> masked attention -> output 1
Head 2 -> masked attention -> output 2
...
Head h -> masked attention -> output h



The head outputs are concatenated and projected using W_O .

15. Self-Attention vs Cross-Attention

Block	Q	K	V	Mask
Decoder self-attention	Decoder	Decoder	Decoder	Causal + target padding
Decoder cross-attention	Decoder	Encoder	Encoder	Encoder padding

Causal masking is required for decoder self-attention because the target is generated autoregressively. In cross-attention, K and V come from the encoder source sequence, so a causal target mask is not normally used. The encoder padding mask blocks source PAD positions.

16. Complete End-to-End Flow

```
Decoder Input
|
Embedding + Positional Encoding
|
Create Q, K, V
|
Split into heads
|
 $QK^T / \sqrt{d_k}$ 
|
Causal Mask x Padding Mask
|
1 -> 0
0 -> -infinity
|
Add mask to scores
|
Softmax
|
Attention weights
|
multiply by V
|
Each head output
|
Concatenate
|
 $W_O$ 
|
Masked Multi-Head Self-Attention Output
```

17. Key Revision Table

Concept	Remember
Causal mask	Stops future target-token visibility.
Padding mask	Stops attention to PAD positions.
Binary convention	1 = allowed; 0 = blocked.
Combining	$\text{FinalMask} = \text{CausalMask} \times \text{PaddingMask}$, element-wise.
Conversion	1 -> 0; 0 -> negative infinity.

Concept	Remember
Location	Add converted mask to scaled QK^T scores before Softmax.
Effect	Blocked positions receive zero Softmax probability.
V	V is weighted by attention probabilities; it is not deleted.

18. Interview Answer

Masked multi-head self-attention is the first attention layer of the Transformer decoder. Since it is self-attention, Q, K and V all come from the decoder input. We calculate scaled dot-product attention independently for each head. A causal mask prevents a position from seeing future target tokens, while a padding mask prevents attention to PAD tokens. If both are binary masks, with 1 meaning allowed and 0 meaning blocked, they can be combined element-wise by multiplication. The resulting 1s are converted to 0 and 0s to negative infinity, and the mask is added to the scaled attention scores before Softmax. Therefore, blocked positions receive zero probability. Finally, the attention weights are multiplied by V, the head outputs are concatenated, and an output projection is applied.

19. One-Line Memory Trick

Causal mask = TIME: don't look ahead.

Padding mask = VALIDITY: don't look at PAD.

Multiply: only positions allowed by both remain 1.

Convert: 1 -> 0, 0 -> negative infinity.

Apply: add to scores -> Softmax -> multiply by V.